

**STATE OF CALIFORNIA
REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL COAST REGION
895 Aerovista Place, Suite 101
San Luis Obispo, CA 93401-7906**

WASTE DISCHARGE REQUIREMENTS ORDER NO. R3-2004-0122

NPDES PERMIT NO. CA0048143

Waste Discharger Identification No. 3 420108001
Proposed for Consideration at the October 22, 2004 Meeting

FOR

**CITY OF SANTA BARBARA
EL ESTERO WASTEWATER TREATMENT FACILITY
SANTA BARBARA COUNTY**

The California Regional Water Quality Control Board, Central Coast Region, (hereafter Regional Board), finds that:

FACILITY OWNER AND LOCATION

1. The City of Santa Barbara (hereafter City Permittee, or Discharger) owns and operates a wastewater collection, treatment, and disposal system (a Publicly Owned Treatment Works, or POTW) to provide sewerage service to the City of Santa Barbara and portions of Santa Barbara County, serving a population of approximately 96,000.
2. The City of Santa Barbara El Estero Wastewater Treatment Facility is located at 520 East Yanonali Street, Santa Barbara, California 93103, on City-owned property in Santa Barbara County (T4N, R27W, Section 23, SB B&M), as shown on Attachment A of this Order.

PURPOSE OF ORDER

3. On March 15, 2004, the Permittee submitted a complete application for authorization to discharge wastes under the National Pollutant Discharge Elimination System (NPDES). The Regional Board last modified NPDES Permit No. CA0048143 on September 8, 1999 (Order No. 99-40).

In accordance with Title 40 of the Code of Federal Regulations (40 CFR), Section 122.46, the term of a NPDES permit may not exceed five years. Order No. 99-40 expired on September 8, 2004. Revisions to Order No. 99-40 are intended to renew the term of NPDES Permit No. CA0048143, and to include all current guidance and regulations applicable to the Discharger's wastewater collection, treatment, and disposal system, the Discharger's wastewater discharge, and to waters receiving treated wastewater from the Discharger.

**TRIBUTARY WASTEWATER COLLECTION
SYSTEM AGENCIES**

4. Santa Barbara County retains ownership and direct responsibility for wastewater collection and transport systems up to the point of discharge into interceptors owned and operated by the City. It is incumbent upon this local wastewater collection entity (as building permit authority) to protect the environment to the greatest degree possible and insure its local collection systems, as well as the receiving sewerage system, are protected and utilized properly. This responsibility includes preventing overflows, and may include restricting or prohibiting the volume, type, or concentration of wastes added to the system.

Staff intends to recommend the regulation of all appropriate tributary wastewater collection agencies under *Waste Discharge Requirements Order No. R3-2004-0123 for Local Wastewater Collection Agencies Tributary to the City of Santa Barbara El Estero Wastewater Treatment Facility, Santa Barbara County*.

The County of Santa Barbara owns five miles of mainlines in Mission Canyon that are maintained by the City. The City and County are signatories to a Memorandum of Understanding regarding the operation and maintenance of this system.

FACILITY OPERATIONAL DESCRIPTIONS

Treatment Processes

5. Secondary treatment of domestic and industrial wastewater consists of screening and grinding, aerated grit removal, primary sedimentation, activated sludge stabilization, secondary clarification, disinfection by chlorination, and dechlorination facilities.

Solid wastes (biosolids) are treated using gravity thickening, dissolved air flotation thickening, anaerobic digestion, and belt press dewatering. Dewatered biosolids are transported and land-applied at various reuse sites or incorporated into processed compost for bulk sale under permit by the appropriate Regional Board and Environmental Health Service.

Design Treatment Capacities

6. The design average dry-weather flow treatment capacity is 11 million gallons per day (MGD), with a design peak flow of 19 MGD. The average influent flow for January to December 2003 was 8.47 MGD (76% of the facility's respective design capacity), and the average dry weather peak flow was 13.13 MGD (69% of the facility's respective design capacity). The average daily flow for the three-year period preceding the Report of Waste Discharge was 8.5 MGD.
7. In accordance with the State Board's Administrative Procedure Manual, Chapter 1,

Publicly Owned Treatment Works (POTWs) with design flows equal to or greater than 1.0 MGD, and those with design flows less than 1.0 MGD but with actual or potential adverse environmental impacts, are classified as major dischargers. As the design flow of the City's facility is 11 MGD, the Regional Board and the U.S. Environmental Protection Agency (USEPA) classify this discharge as a major discharge.

Biosolids Disposal

8. On February 19, 1993, the USEPA issued the final rule for the use and disposal of biosolids (40 CFR Part 503), a regulation that governs the final use or disposal of biosolids. The intent of this Federal program is to ensure that biosolids are used or disposed of in a way that protects both human health and the environment.

The promulgated regulations require that producers of biosolids meet certain reporting, handling, and disposal requirements. As the USEPA has not delegated the authority to implement the biosolids program to the State of California, the enforcement of biosolids requirements applying to the Permittee remains under USEPA's jurisdiction at this time. USEPA, not this Regional Board, will oversee compliance with 40 CFR Part 503. See Section F, *Biosolids Requirements*, and MRP Section VIII, *Biosolids Monitoring, Reporting, and Notification*.

9. Biosolids generated by the Permittee are both land-applied and sold in bulk as a component of processed compost under permit by the Regional Boards and County Environmental Health Services with jurisdiction. In 2003, biosolids were: 1) Land-applied at McCarthy Family Farms, Inc. in Kern County under permits from the Central Valley Regional Board (Order No. R5-2002-0172) and the Kern County Environmental Health Service, and; 2) Incorporated into processed compost and sold in bulk by Engel & Gray in Santa Barbara County under permits by the Central Coast Regional Board (Order No. 99-11) and the Santa Barbara County Environmental Health Service.

Domestic Septage Receiving

10. Domestic septage is liquid and/or solid waste material that has been removed from a septic tank, portable toilet, Type III marine sanitation device, or similar source that receives only domestic sewage. The City's facility has a dedicated septage receiving station for small-volume waste haulers. All mobile waste haulers are licensed and permitted by the City of Santa Barbara.

Wastewater Reclamation

11. The facility provides tertiary wastewater treatment by means of coagulation, flocculation, filtration, and additional disinfection processes. The additional treatment allows the Discharger to provide up to 4.3 MGD of reclaimed wastewater for irrigation at locations throughout the City, and for toilet flushing at restrooms in many of the City's park restrooms. WDRs and Master Reclamation Permit Order No. 97-44 governs the use of the reclaimed wastewater in accordance with the wastewater reclamation criteria specified in Title 22 of the California Code of Regulations.

Brine Wastewater Disposal

12. The City owns a seawater desalination facility, which is currently deactivated until needed. When operational, the desalination plant discharges waste brine at one of five flowrates depending on the facility's rate of freshwater production. The waste brine discharge flowrates are 3.9, 4.1, 9.4, 10, and 12.5 MGD. Due to its high salinity, the brine is substantially denser than the ambient ocean waters. As the fraction of brine in the combined brine/POTW discharge increases, the combined discharge becomes less buoyant.
13. As estimated by computer modeling, the following table provides: (1) the minimum WWTP discharge flowrate necessary to ensure the combined discharge will remain buoyant and above the seafloor, and; (2) the minimum initial dilution ratio (MIDR) for the combined discharge computed at the minimum POTW discharge flowrate.

Brine Discharge, MGD	WWTP Discharge, MGD	MIDR
3.9	5	55
4.1	4	44
9.4	8	52
10	10	56
12.5	14	62

The City does not currently accept waste brine from other sources.

Storm Water Management

14. Storm water runoff due to rainfall which falls upon the wastewater treatment facility and which may be exposed to on-site pollutant sources is routed to the facility's headworks for treatment. This permit regulates all storm water discharges at this facility and complies with Federal regulations for storm water management [Title 40, Code of Federal Regulations (CFR), Parts 122, 123, and 124].

Ocean Outfall

15. Secondary treated municipal wastewater is discharged to the Pacific Ocean through the Discharger's 8,720-foot outfall/diffuser system. The outfall (34° 23' 31" N. Latitude, 119° 40' 08" W. Longitude) terminates in the Santa Barbara Channel in approximately 70 feet of water. The minimum initial dilution ratio of the outfall / diffuser system is 120:1 (seawater:effluent, or parts seawater to parts effluent) without desalination facility brine discharge, and 44:1 with brine discharge. The hydraulic capacity of the outfall is 28 MGD. The outfall location is shown on Attachment A.

CALIFORNIA OCEAN PLAN

16. The State Water Resources Control Board (State Board) adopted the *Water Quality Control Plan - Ocean Waters of California* (Ocean Plan) on November 16, 2000. U.S. Environmental Protection Agency (USEPA) subsequently approved the Ocean Plan on December 3, 2001. The Ocean Plan contains water quality

objectives and other requirements governing discharge of wastes to the Pacific Ocean, and is applicable to the waste discharge regulated by this Order.

17. In accordance with the Ocean Plan, this Order specifies numeric effluent limitations in terms of concentration and mass emission rate.

BASIN PLAN

18. The *Water Quality Control Plan, Central Coastal Region* (Basin Plan), was revised and adopted by the Regional Board on September 8, 1994. The Basin Plan incorporates statewide plans and policies by reference and contains a strategy for protecting beneficial uses of State waters.

19. Existing and anticipated beneficial uses of the Pacific Ocean in the vicinity of the marine discharge include:

- a. Water contact recreation (REC-1);
- b. Non-contact water recreation (REC-2);
- c. Industrial service supply (IND);
- d. Navigation (NAV);
- e. Marine habitat (MAR);
- f. Shellfish harvesting (SHELL);
- g. Commercial and sport fishing (COMM);
- h. Rare, threatened, or endangered species (RARE);
- i. Wildlife habitat (WILD);
- j. Aesthetic enjoyment;
- k. Mariculture or Aquaculture (AQUA);
- l. Fish migration (MIGR), and;
- m. Fish spawning (SPWN).

20. The shellfish beneficial use (Finding 19.f) exists wherever mussels, clams, or oysters may be harvested for human consumption. To the knowledge of this Regional Board: 1) habitat for mussels is very limited, existing only at shoreline locations and offshore oil platforms greater than 1-1/2 miles from the discharge, and; 2) clamming activity is insignificant.

21. The State Department of Health Services has established a prohibitive zone for shellfish harvesting within a three-mile radius of the discharge. Thus, shellfish harvesting is an

existing beneficial use in nearshore areas (i.e., within one mile of shore) and outside the three-mile prohibitive zone, and receiving water limitations specified in paragraph C.1.b of this Order apply in these areas.

22. The Discharger is required to notify the Regional Board, Department of Health Services, and any Mariculture Grower as soon as possible when there is a loss of disinfection or if three consecutive total effluent coliform bacteria tests exceed 16,000 per 100 mL.

The Discharger also has a voluntary agreement (*Memorandum of Understanding* (MOU) *By and Between the California Department of Health Services and City of Santa Barbara*, executed January 31, 2001) with the California Department of Health Services regarding the notification of all interested parties when a significant breakdown in disinfection capabilities occurs at the facility. Order and MRP No. R3-2004-0122 contain the same language as Order and MRP No. 99-40 contained with respect to the MOU (see MRP Table 2, Footnote 3).

FEDERAL CLEAN WATER ACT

23. The Federal Clean Water Act (CWA) is codified in Title 33 of the United States Code (33 USC) commencing with Section 1251. Water quality standards, plans, guidelines, standards of performance, effluent limitations, toxic and industrial pretreatment standards, biosolids or sludge use and disposal regulations, ocean discharge criteria, and enforcement authority established under CWA Sections [parallel 33 USC references bracketed] 301 [1311], 302 [1312], 303(d) [1313], 304 [1314], 306 [1316], 307 [1317], 308 [1318], 309 [1319], 402 [1342], 403 [1343], and 405 [1345], as amended, are applicable to the discharge permitted by this Order.

CALIFORNIA WATER CODE

24. The Porter-Cologne Water Quality Control Act (Porter-Cologne) is codified in Division 7 of the California Water Code and commences at Section 13000. Porter-Cologne establishes the

legal authority by which the State and Regional Boards implement water quality control, monitoring, reporting, and enforcement measures to protect all State waters.

THE CLEAN WATER ENFORCEMENT AND POLLUTION PREVENTION ACT OF 1999

25. Effective January 1, 2000, the Clean Water Enforcement and Pollution Prevention Act of 1999 (Act), amended California Water Code Section 13385. The Act requires the Regional Board to impose mandatory minimum penalties for certain violations. Failure to comply with NPDES Permit effluent limitations and certain other requirements and conditions contained in this Order may result in significant and mandatory enforcement action by the Regional Board.

MONITORING AND REPORTING PROGRAM

26. The Discharger is required to comply with Monitoring and Reporting Program No. R3-2004-0122, which is part of this Order.

PRETREATMENT PROGRAM FOR INDUSTRIAL WASTEWATER

27. Title 40 of the Code of Federal Regulations, Section 403 (40 CFR Part 403) establishes pretreatment program requirements for POTWs which receive pollutants from industries subject to pretreatment standards. This Order contains industrial pretreatment program requirements pursuant to 40 CFR Part 403. (See *Pretreatment Specifications*, Section E).
28. 40 CFR 403 establishes that industrial pretreatment programs are mandatory for POTWs with design flow rates equal to or greater than 5 MGD. A pretreatment program is therefore mandatory for this facility (design flow rate is 11 MGD). The Discharger implements an industrial pretreatment program originally approved by USEPA in 1983. Discharges from industrial sources are not known to have interfered with treatment plant operation or contributed to effluent violations. The discharger will continue to implement its

existing pretreatment program. See Section E, *Pretreatment Specifications*.

WASTEWATER COLLECTION SYSTEM OVERFLOWS

29. The Discharger's sanitary sewer (wastewater collection) system collects wastewater using pipes, pumps, and/or other conveyance systems, and directs the raw sewage to the wastewater treatment facility. A "wastewater collection system overflow" is defined as a discharge to ground or surface water from the wastewater collection system at any point upstream of the wastewater treatment facility. Temporary storage and conveyance facilities (such as wet wells, regulated impoundments, tanks, highlines, etc.) may be part of a wastewater collection system, and discharges to these facilities are not considered wastewater collection system overflows provided that the waste is fully contained within these temporary storage/conveyance facilities.
30. Wastewater collection system overflows consist of varying mixtures of domestic sewage, industrial wastewater, and commercial wastewater, the mixture depending upon the pattern of land use in the wastewater collection system tributary to an overflow location. The chief causes of wastewater collection system overflows include, but are not limited to, line blockages due to grease, roots, or debris, sewer line flood damage, manhole structure failures, vandalism, pump station mechanical failures, power outages, storm or ground water inflow/infiltration, lack of capacity, and contractor-related incidents.
31. Wastewater collection system overflows often contain high levels of suspended solids, pathogenic organisms, toxic pollutants, nutrients, oxygen demanding organic compounds, oil and grease, and other pollutants. Wastewater collection system overflows can pose a threat to public health, cause temporary exceedances of applicable water quality objectives, adversely affect aquatic life, and impair the public recreational use and aesthetic enjoyment of surface waters in the area.

32. The Discharger is expected to take all reasonably necessary steps to adequately operate and maintain its wastewater collection system to prevent overflows. This Order requires that the Discharger continue to implement and update its Collection System Maintenance and Renovation Program, and further requires the development of a Wastewater Collection System Management Plan (see Section D, *Wastewater Collection System Requirements*, of this Order, and Attachment 1 to the MRP).
33. This Order requires the Discharger to report wastewater collection system overflows in accordance with MRP No. R3-2004-0122, Section XII, *Wastewater Collection System Spill/Overflow Reporting*.

ANTI-DEGRADATION POLICY

34. Waste discharge requirements for this discharge must be in conformance with 40 CFR 131.12 and State Board Resolution No. 68-16, *Statement of Policy with Respect to Maintaining High Quality of Waters in California* (known collectively as "anti-degradation" policies). These policies are intended to maintain and protect the existing beneficial uses of receiving waters and the levels of water quality necessary to achieve those goals. The Regional Board has taken into consideration the requirements of the State and Federal anti-degradation policies in establishing the requirements contained herein, and has determined that any reduction in water quality as a result of this discharge will not result in any long-term deleterious effects on water quality or associated beneficial uses.

ANTI-BACKSLIDING POLICY

35. 40 CFR 122.44(l) requires effluent limitations for reissued NPDES permits at least as stringent as the previous permit, with some exceptions. As the effluent limitations, standards, or conditions in this Order are the same as or more stringent than those in Order No. 99-40 (except for differences due to rounding, significant figures, or undetected calculation errors), adoption of this Order is consistent with anti-backsliding policies.

REASONABLE POTENTIAL ANALYSIS

36. Federal regulations governing the Federal and State NPDES permit program require that NPDES permits contain effluent limitations for all pollutant parameters that, "...may be discharged at a level which will cause, have the reasonable potential to cause, or contribute to an excursion above any State water quality standard, including State narrative criteria for water quality." [40 CFR 122.44 (d)] The Ocean Plan specifies numeric water quality objectives for the constituents specified in the Effluent Monitoring Section of Monitoring and Reporting Program No. R3-2004-0122. No statistical reasonable potential analysis was conducted on the discharge. Based on the discussion presented in the Staff Report for this Order and the nature of the facility, the discharge warrants the specification of effluent limits based on all of the Ocean Plan water quality objectives. These effluent limits are shown in Section B of this Order.

CALIFORNIA ENVIRONMENTAL QUALITY ACT (CEQA)

37. The issuance of waste discharge requirements for this discharge is exempt from the requirement for preparation of environmental documents under the California Environmental Quality Act (Public Resources Code, Division 13, Chapter 3, commencing with Section 21000) in accordance with California Water Code Section 13389, and 14 California Code of Regulations Section 15301 (existing facilities).

GENERAL FINDINGS

38. A permit and the privilege to discharge waste into waters of the State is conditional upon the discharge complying with provisions of Division 7 of the California Water Code and of the Clean Water Act (as amended or supplemented by implementing guidelines and regulations), and with any more stringent effluent limitations necessary to implement water quality control plans, to protect beneficial uses, and to prevent nuisance.

39. On September 8, 2000, the Governor of California approved AB2800, which added sections to the Public Resources Code that are relevant to Areas of Special Biological Significance. Effective January 1, 2003, Section 36700(f) of the Public Resources Code named Areas of Special Biological Significance (ASBS) as State Water Quality Protection Areas (SWQPA).

The Ocean Plan prohibits the discharge of waste to designated ASBS except as provided in the Ocean Plan, Chapter III, Section E, *Implementation Provisions for ASBS*. Discharges shall be located a sufficient distance from such designated areas to assure maintenance of natural water quality conditions in these areas. ASBS are those areas designated by the State Water Resources Control Board (SWRCB) as requiring protection of species or biological communities to the extent that alteration of natural water quality is undesirable. ASBS are designated by the SWRCB following the procedures provided in Appendix IV of the Ocean Plan. See Appendix V of the Ocean Plan for ASBS designated at the time of this Order's issuance, and subsequent revised listings established by the SWRCB for either ASBS or SWQPA.

The City does not discharge waste to ASBS, nor does staff have any information indicating that the discharge location is being considered for ASBS designation.

40. This Order shall serve as a National Pollutant Discharge Elimination System (NPDES) Permit for the City of Santa Barbara waste discharge through the City's ocean outfall/diffuser system to the Pacific Ocean pursuant to Section 402 of the Clean Water Act, and amendments thereto. Compliance with this Order should mitigate any potential changes in water quality resulting from the permittee's discharge of waste.
41. Any person affected by this action of the Regional Board may petition the State Water Resources Control Board (State Board) to review the action in accordance with Section 13320 of the California Water Code and Title 23, California Code of Regulations, Section

2050. The petition must be received by the State Board within 30 days of the adoption date of this Order. Copies of the law and regulations applicable to filing petitions are available at http://www.swrcb.ca.gov/water_laws/cawtrcode/wqpetition_instr.html, or will be provided upon request.

42. On July 16, 2004, the Regional Board notified the Discharger and all known interested parties of its intent to renew the NPDES permit/waste discharge requirements for waste discharges from the City of Santa Barbara wastewater treatment facility, through the City of Santa Barbara Ocean Outfall, to the Pacific Ocean. The Regional Board's notification included a scheduled public hearing date, and provided interested parties with a copy of the proposed Order and an opportunity to submit written comments.
43. In a public hearing on October 22, 2004, the Regional Board heard and considered all comments pertaining to the City of Santa Barbara's waste discharge to the Pacific Ocean, and found this Order consistent with the above findings.

IT IS HEREBY ORDERED, pursuant to authority in Sections 13263, 13383, 13377, and 13523 of the California Water Code, that the City of Santa Barbara, its agents, successors, and assigns, may discharge waste from the City of Santa Barbara El Estero Wastewater Treatment Facility to the Pacific Ocean providing they comply with the following:

[Permit conditions, definitions, and methods of determining compliance are also contained in the attached *Standard Provisions and Reporting Requirements for National Pollutant Discharge Elimination System Permits* (Standard Provisions), dated January 1985 (See Provision G.3 of this Order). Throughout this Order, terms in **bold** and within quotation marks (" ") are defined in the attached Standard Provisions or the Ocean Plan].

Requirements in this Order are provided with the following superscripts to indicate their origin:

- A Title 40, Code of Federal Regulations, Sections 122 and 133
- B California Ocean Plan
- C Central Coast Water Quality Control Plan (Basin Plan)
- D California Code of Regulations, Title 17, Sections 7957 and 7958

All technical and monitoring reports submitted according to this Order are required pursuant to Sections 13267 and 13383 of the California Water Code. Failure to submit reports in accordance with schedules established by this Order or attachments to this Order, or failure to submit a report of sufficient technical quality to be acceptable to the Executive Officer, may subject the Permittee to enforcement action pursuant to Sections 13268 and 13385 of the California Water Code.

A. DISCHARGE PROHIBITIONS

Please see Standard Provisions, Section A, *General Permit Conditions*, Prohibition Nos. 1-7.

1. The discharge of treated wastewater at locations other than those listed below is prohibited:
 - a. City of Santa Barbara Ocean Outfall Diffuser (34° 23' 31" N. Latitude, 119° 40' 08" W. Longitude); and
 - b. Approved reclaimed water use sites authorized under valid water reclamation requirements issued or waived by the Regional Board.

B. DISCHARGE SPECIFICATIONS

1. Effluent daily dry-weather flow shall not exceed a monthly average of 11 MGD.
2. Effluent shall not exceed the limitations specified in Tables A and B-1 through B-3, which apply to the wastewater effluent, or Attachment B, Tables B-1b through B-3b for wastewater effluent with desalination facility brine wastewater discharged from the City of Santa Barbara El Estero Wastewater Treatment Facility through the City of Santa Barbara Ocean Outfall to the Pacific Ocean^B.

The Ocean Plan states that waste discharge requirements shall also specify effluent limitations in terms of mass emission rate limits utilizing the general formula:

$$\text{lbs/day} = 0.00834 \times C_e \times Q$$

where: C_e = the effluent concentration limit, in $\mu\text{g/L}$, and

Q = the flow rate observed over the concentration limit's period (e.g., daily, weekly, monthly/30-day, 6-month), in millions of gallons per day (MGD)

Note: If C_e expressed in units of mg/L , use a conversion factor of 8.34 instead of 0.00834.

This formula applies to Table A effluent limits for Five-Day Carbonaceous Biochemical Oxygen Demand (CBOD_5), Total Suspended Solids (TSS), and Oil & Grease, and to Tables B-1 through B-3 (or Attachment B, Tables B-1b through B-3b for discharges which include desalination facility waste brine).

3. For daily dry weather flows equal to or less than a monthly average of 11 MGD, the effluent mass emission rate shall not exceed the "Maximum Allowable Mass Emission Rate." The "Maximum

Allowable Mass Emission Rate," whether for a month, week, day, or six-month period, is a daily rate determined with the formula in Discharge Specification B.2 using the effluent concentration limit specified in this permit for the period and the average of measured daily flows (up to the allowable flow) over the period (see Standard Provisions G.11 - 13).

4. The monthly/30-day average percent removal ("**Removal Efficiency**") for CBOD₅ and TSS shall not be less than 85 percent ^A.
5. The median number of total coliform organisms in effluent shall not exceed 2,300 per 100 milliliters (mL), as determined by the bacteriological results for the last 7 days on which analyses were completed, and the number of total coliform organisms in any sample shall not exceed 16,000 per 100 mL.
6. The median number of fecal coliform organisms in effluent shall not exceed 460 per 100 mL, as determined by the bacteriological results for the last 7 days on which analyses were completed, and the number of fecal coliform organisms in any sample shall not exceed 3,200 per 100 mL.

TABLE A – Effluent Limitations for Major Constituents and Properties of Wastewater

Constituents	Unit of Measurement	Monthly (30-Day) Average	Weekly (7-Day) Average	Daily Maximum
CBOD ₅ ^A	mg/L	25 ^A	40 ^A	90
Total Suspended Solids	mg/L	30 ^A	45 ^A	90
Oil & Grease ^B	mg/L	25	40	75
Settleable Solids ^B	mL/L	1.0	1.5	3.0
Turbidity ^B	NTU	75	100	225
pH ^{A, B}	pH units	Between 6.0 and 9.0 at all times		

^A Per 40 CFR 133.102, this Order substitutes five-day Carbonaceous Biochemical Oxygen Demand (CBOD₅) effluent limits for five-day Biochemical Oxygen Demand (BOD₅) effluent limits.

7. When the discharge consists of municipal wastewater only, without the brine wastewater discharge from the desalination facility, effluent shall not exceed the effluent limitations listed in Tables B-1 through B-3 (effluent limitations derived using a minimum initial seawater-to-effluent dilution ratio of 120 to 1).

When the discharge consists of municipal wastewater with brine wastewater from the desalination facility, effluent shall not exceed the effluent limitations listed in Tables B-1b through B-3b of Attachment B to this Order (effluent limitations derived using a minimum initial seawater-to-effluent dilution ratio of 44 to 1).

TABLE B-1 – Effluent Limitations for the Protection of Marine Aquatic Life

	Units of Measurement	6-Month Median ⁴	Daily Maximum ⁵	Instantaneous Maximum ⁶
Arsenic	ug/L	610	3,500	9,300
Cadmium	ug/L	120	490	1,200
Chromium (Hexavalent) ¹	ug/L	240	970	2,400
Copper	ug/L	120	1,200	3,400

**TABLE B-1 – Effluent Limitations for the Protection of
Marine Aquatic Life**

	Units of Measurement	6-Month Median ⁴	Daily Maximum ⁵	Instantaneous Maximum ⁶
Lead	ug/L	240	970	2,400
Mercury	ug/L	4.8	19	48
Nickel	ug/L	600	2,400	6,000
Selenium	ug/L	1,800	7,300	18,000
Silver	ug/L	66	320	830
Zinc	ug/L	1,500	8,700	23,000
Cyanide ²	ug/L	120	480	1,200
Total Chlorine Residual ³	ug/L	240	970	7,300
Ammonia (expressed as N)	ug/L	73,000	290,000	730,000
Acute Toxicity	TUa	-----	3.9	-----
Chronic Toxicity	TUc	-----	121	-----
Phenolic Compounds (non-chlorinated)	ug/L	3,600	14,000	36,000
Chlorinated Phenolics	ug/L	120	480	1,200
Endosulfan	ug/L	1.1	2.2	3.3
Endrin	ug/L	0.24	0.48	0.73
HCH	ug/L	0.48	0.97	1.4
Radioactivity	Not to exceed limits specified in Title 17, Division 1, Chapter 5, Subchapter 4, Group 3, Article 3, Section 30253 of the California Code of Regulations. Reference to Section 30253 is prospective, including future changes to any incorporated provisions of federal law, as the changes take effect.			

¹ Dischargers may, at their option, meet this limitation as a total chromium limitation. ^B

² If a Discharger can demonstrate to the satisfaction of the Regional Board (subject to USEPA approval) that an analytical method is available to reliably distinguish between strongly and weakly complexed cyanide, effluent limitations for cyanide may be met by the combined measurement of free cyanide, simple alkali metal cyanides, and weakly complexed organometallic cyanide complexes. In order for the analytical method to be acceptable, the recovery of free cyanide from metal complexes must be comparable to that achieved by the approved method in 40 CFR PART 136, as revised July 1, 2003, or later. ^B

³ Water quality objectives for total chlorine residual applying to intermittent discharges not exceeding two hours shall be determined using the following equation:

$$\log y = -0.43 (\log x) + 1.8$$

where: y = the water quality objective (in ug/L) to apply when chlorine is **being discharged**; and
x = the duration of uninterrupted chlorine discharge in minutes.

The applicable effluent limitation must then be determined using Equation No. 1 from the Ocean Plan. ^B

⁴ The six-month median shall apply as a moving median of daily values for any 180-day period in which daily values represent flow weighted average concentrations within a 24-hour period. For intermittent discharges, the daily value shall be considered to equal zero for days on which no discharge occurred. The six-month median limit on daily mass emissions shall be determined using the six-month median effluent concentration as Ce and the observed flow rate Q in millions of gallons per day (each variable referring to Equation 3 of the Ocean Plan). ^B

⁵ The daily maximum shall apply to flow weighted 24-hour composite samples. The daily maximum mass emission shall be determined using the daily maximum effluent concentration limit as C_e and the observed flow rate Q in millions of gallons per day (each variable referring to Equation 3 of the Ocean Plan). ^B

⁶ The instantaneous maximum shall apply to grab sample determinations. ^B

TABLE B-2 – Effluent Limitations for the Protection Of Human Health – Non-Carcinogens

Chemical	Units of Measurement	30-day average
Acrolein	ug/L	2.7×10^4
Antimony	ug/L	1.4×10^5
Bis(2-chloroethoxy) methane	ug/L	530
Bis(2-chloroisopropyl) ether	ug/L	1.4×10^5
Chlorobenzene	ug/L	6.9×10^4
Chromium (III)	ug/L	2.3×10^7
di-n-butyl phthalate	ug/L	4.2×10^5
Dichlorobenzenes	ug/L	6.2×10^5
Diethyl phthalate	ug/L	4.0×10^6
Dimethyl phthalate	ug/L	9.9×10^7
4,6-dinitro-2-methylphenol	ug/L	2.7×10^4
2,4-dinitrophenol	ug/L	480
Ethylbenzene	ug/L	5.0×10^5
Fluoranthene	ug/L	1.8×10^3
Hexachlorocyclopentadiene	ug/L	7.0×10^3
Nitrobenzene	ug/L	590
Thallium	ug/L	240
Toluene	ug/L	10×10^6
Tributyltin	ug/L	0.17
1,1,1-trichloroethane	ug/L	6.5×10^7

TABLE B-3 – Effluent Limitations for the Protection Of Human Health – Carcinogens

Chemical	Units of Measurement	30-day average
Acrylonitrile	ug/L	12
Aldrin	ug/L	2.7×10^{-3}
Benzene	ug/L	710
Benzidine	ug/L	8.4×10^{-3}
Beryllium	ug/L	4.0
Bis(2-chloroethyl) ether	ug/L	5.4
Bis(2-ethylhexyl) phthalate	ug/L	420
Carbon tetrachloride	ug/L	110
Chlordane	ug/L	2.8×10^{-3}
Chlorodibromomethane	ug/L	1.0×10^3
Chloroform	ug/L	1.6×10^4
DDT	ug/L	0.021

**TABLE B-3 – Effluent Limitations for the Protection Of Human Health –
Carcinogens**

Chemical	Units of Measurement	30-day average
1,4-dichlorobenzene	ug/L	2.2×10^3
3,3'-dichlorobenzidine	ug/L	0.98
1,2-dichloroethane	ug/L	3.4×10^3
1,1-dichloroethylene	ug/L	110
Dichlorobromomethane	ug/L	750
Dichloromethane	ug/L	5.4×10^4
1,3-dichloropropene	ug/L	1.1×10^3
Dieldrin	ug/L	4.8×10^{-3}
2,4-dinitrotoluene	ug/L	310
1,2-diphenylhydrazine	ug/L	19
Halomethanes	ug/L	1.6×10^4
Heptachlor	ug/L	6.0×10^{-3}
Heptachlor epoxide	ug/L	2.4×10^{-3}
Hexachlorobenzene	ug/L	0.025
Hexachlorobutadiene	ug/L	1.7×10^3
Hexachloroethane	ug/L	300
Isophorone	ug/L	8.8×10^4
N-nitrosodimethylamine	ug/L	880
N-nitrosodi-N-propylamine	ug/L	46
N-nitrosodiphenylamine	ug/L	300
PAHs	ug/L	1.1
PCBs	ug/L	2.3×10^{-3}
TCDD equivalents	ug/L	4.7×10^{-7}
1,1,2,2-tetrachloroethane	ug/L	280
Tetrachloroethylene	ug/L	240
Toxaphene	ug/L	0.025
Trichloroethylene	ug/L	3.3×10^3
1,1,2-trichloroethane	ug/L	1.1×10^3
2,4,6-trichlorophenol	ug/L	35
Vinyl chloride	ug/L	4.4×10^3

8. Waste discharged to the Ocean must be essentially free of ^B:
 - a. Material that is floatable or will become floatable upon discharge;
 - b. Settleable material or substances that may form sediments which will degrade benthic communities or other aquatic life;
 - c. Substances which will accumulate to toxic levels in marine waters, sediments, or biota;
 - d. Substances that significantly decrease the natural light to benthic communities and other marine life; and
 - e. Materials that result in aesthetically undesirable discoloration of the ocean surface.
9. Effluent limitations derived from Ocean Plan Tables A and B (i.e., the effluent limitations presented in Tables A, B-1 through B-3, and Attachment B Tables B-1b through B-3b of this Order) shall apply to the Discharger's total effluent, of whatever origin (i.e., gross, not net, discharge), except where otherwise specified in the Ocean Plan ^B.
10. The discharge of waste shall not cause water quality objectives established in the California Ocean Plan, Table B, to be exceeded in the receiving water upon completion of initial dilution, except that objectives indicated for radioactivity shall apply directly to the undiluted waste effluent ^B.
11. The effluent limitations of this Order are based on California Ocean Plan criteria and equations as applicable therein, using a minimum initial dilution of 120:1 (seawater:effluent) for effluent discharges without desalination facility brine, and 44:1 for discharges with brine. If the actual dilution ratio for either circumstance is found to be lower or higher, then the ratio will be recalculated and this Order revised when and as appropriate.
12. Effluent discharged to the Pacific Ocean shall encounter the seafloor only after the seawater to

effluent dilution ratio has increased to the minimum ratio specified in Effluent Limitation Tables B-1 through B-3 (or Tables B-1b through B-3b of Attachment B to this Order), or as determined according to Provisions G.6 and G.7. The dilution ratio shall be demonstrated by means of a computer model approved by the Executive Officer, employing input variables approved by the Executive Officer.

13. The minimum initial dilution is the lowest average initial dilution within any single month of the year. Dilution estimates shall be based on observed waste flow characteristics, observed receiving water density structure, and the assumption that no currents (of sufficient strength to influence the initial dilution process) flow across the discharge structure ^B.
14. The State Board shall identify standard dilution models for use in determining the minimum initial dilution, and shall assist the Regional Board in its evaluation for specific waste discharges. Dischargers may propose alternative methods of calculating minimum initial dilution, and the Regional Board may accept such methods upon verification of its accuracy and applicability ^B.
15. If only one sample is collected during the time period associated with an effluent limitation or water quality objective (e.g., 30-day average or 6-month median), the single measurement shall be used to determine compliance with the effluent limitation for the entire time period ^B.
16. Any significant change in waste flow shall be cause for reevaluating effluent limitations ^B.

C. RECEIVING WATER LIMITATIONS ^{B,D}

[Receiving water quality is a result of many factors, some unrelated to the discharge. This permit considers these factors and is designed to minimize the influence of the discharge to the receiving water. Compliance with Receiving Water Limitations shall be determined from samples collected at stations representative of the area of potential influence but outside the zone of initial dilution.]

At the time of this Order's consideration for adoption, the State Board proposed revisions to the Ocean Plan which may significantly affect Sections C.1 and C.2 below, and MRP Section III. The Executive Officer will formally notify the Discharger of any applicable Ocean Plan changes. The Regional Board may defer the formal revision of this Order and MRP until the next scheduled renewal if permitted by the Ocean Plan, or may reopen the Order to amend it to comply with the Ocean Plan revisions.

1. Bacterial Characteristics

Discharge shall not cause the following water quality objectives to be violated in ocean waters upon completion of "initial dilution":

- a. Body-Contact Standards – Within a zone bounded by the shoreline and a distance of 1,000 feet from the shoreline or the 30-foot depth contour, whichever is further from the shoreline, and in areas outside this zone used for body-contact sports, as determined by the Regional Board, but including all kelp beds, the following bacteriological objectives shall be maintained throughout the water column:

- 1) Samples of water from each sampling station shall have a density of total coliform organisms less than 1,000 per 100 mL (10 per mL); provided that not more than 20 percent of samples at any sampling station, in any 30-day period, may exceed 1,000 per 100 mL (10 per mL), and provided further that no single sample when verified by a repeat sample taken within 48 hours shall exceed 10,000 per 100 mL (100 per mL). Furthermore, if the ratio of fecal to total coliform in a single sample exceeds 0.1, the density of total coliform organisms shall not exceed 1,000 per 100 mL.
- 2) The fecal coliform density, based on a minimum of not less than five samples for any 30-day period, shall not exceed a "geometric mean" of 200 per 100 mL, nor shall more than ten percent of

the total samples during any 60-day period exceed 400 per 100 mL.

- 3) The enterococcus density, based on a single sample, shall not exceed 104 per 100 mL, nor shall the geometric mean, based on a minimum of at least five samples from a single sampling station for any 30-day period, exceed 35 per 100 mL.

- b. Shellfish Harvesting Standards – At all areas where "shellfish" may be harvested for human consumption, as determined by the Regional Board (see Findings 19.f and 20-22), the following bacteriological objectives shall be maintained throughout the water column:

- 1) In any 60-day period, the "median" total coliform density shall not exceed 70 per 100 mL, and not more than ten percent of the samples shall exceed 230 per 100 mL.

2. Implementation Provisions for Bacterial Assessment and Remedial Action Requirements

The requirements listed below shall be used to determine the occurrence and extent of any impairment of a beneficial use due to bacterial contamination, generate information which can be used in the development of an enterococcus standard, and provide the basis for remedial actions necessary to minimize or eliminate any impairment of a beneficial use.

- a. Measurement of enterococcus density shall be conducted at all stations where measurement of total and fecal coliforms is required. In addition to the requirements of Receiving Water Limitation C.1, above, if a shore or 30-foot contour sampling station consistently exceeds a coliform objective or exceeds a geometric mean enterococcus density of 24 organisms per 100 ml for a 30-day period, or 12 organisms per 100 ml for a six-month period, the Discharger shall conduct a survey to determine if the discharge is the source of the contamination. The geometric mean shall

be a moving average based on no less than five samples per month, spaced evenly over the time interval. When a sanitary survey identifies a controllable source of indicator organisms associated with a discharge of sewage, the Discharger shall take action to control the source.

- b. The Discharger shall conduct sanitary surveys when so directed by the Regional Board or the Executive Officer. The Discharger shall control any controllable discharges identified in a sanitary survey.

3. Physical Characteristics

- a. Floating particulates and grease and oil shall not be visible.
- b. The discharge of "waste" shall not cause aesthetically undesirable discoloration of the ocean surface.
- c. "Natural light" shall not be "significantly" reduced at any point outside the "zone of initial dilution" as the result of the discharge of "waste".
- d. The rate of deposition of inert solids and the characteristics of inert solids in ocean sediments shall not be changed such that benthic communities are degraded.
- e. Temperature of the receiving water shall not be altered to adversely affect beneficial uses.

4. Chemical Characteristics

- a. The dissolved oxygen concentration shall not at any time be depressed more than 10 percent from that which occurs naturally ^B, or fall below 5.0 mg/L ^C, as the result of the discharge of oxygen demanding "waste" materials. The mean annual dissolved oxygen concentration shall not be less than 7.0 mg/L ^C.
- b. The pH shall not be changed at any time more than 0.2 units from that which occurs

naturally, and shall be within the range of 7.0 to 8.5 at all times.

- c. The dissolved sulfide concentrations of waters in and near sediments shall not be "significantly" increased above that present under natural conditions.
- d. The concentrations of substances set forth in Table B of the Ocean Plan shall not be increased in marine sediments to levels which would "degrade" indigenous biota.
- e. The concentration of organic materials in marine sediments shall not be increased to levels which would "degrade" marine life.
- f. Nutrient materials shall not cause objectionable aquatic growth or "degrade" indigenous biota.

5. Biological Characteristics

- a. Marine communities, including vertebrate, invertebrate, and plant species, shall not be "degraded."
- b. The natural taste, odor, and color of fish, "shellfish," or other marine resources used for human consumption shall not be altered.
- c. The concentration of organic materials in fish, "shellfish", or other marine resources used for human consumption shall not bioaccumulate to levels that are harmful to human health.

6. Radioactivity

- a. Discharge of radioactive "waste" shall not "degrade" marine life.
- b. Radionuclides shall not be present in concentrations that are deleterious to human, plant, animal, or aquatic life; or result in the accumulation of radionuclides in the food web to an extent which presents a hazard to human, plant, animal, or aquatic life. ^C

7. General Standards

- a. The discharge shall not cause deposition of sewage, sludge, grease, or other physical evidence of sewage discharge on beaches, rocks, or shorelines, and material of sewage origin shall not be visible in the water.
- b. The discharge shall not cause a violation of any applicable water quality objective or standard for receiving waters adopted by the Regional Board or the State Board, as required by the Clean Water Act and regulations adopted thereunder.

D. WASTEWATER COLLECTION SYSTEM REQUIREMENTS**Wastewater Collection System Management Plan Development and Implementation**

1. The Permittee shall develop and implement a Wastewater Collection System Management Plan (Management Plan) in accordance with the time schedule established in Section XI of Attachment 1 to MRP No. R3-2004-0122. The Management Plan shall be available to any member of the public upon written request.
2. The Permittee shall provide the Santa Barbara County Public Works Department with a copy of the Management Plan annual report required by this Order.
3. The essential elements of the Management Plan are outlined in Attachment 1 of MRP No. R3-2004-0122. All elements of the Management Plan outlined in MRP Attachment 1 shall be clearly labeled and addressed by the Permittee. If any element of MRP Attachment 1 is not appropriate or applicable to a Permittee's Management Plan, then the Plan shall provide the rationale for not including the element.
4. To facilitate continuity between the Permittee's existing wastewater collection system programs and the development and implementation of the Management Plan, the Plan shall incorporate within the appropriate

Plan sections, but not be limited to, the Permittee's Collection System Maintenance and Renovation Program, and the *Wastewater Collection System Overflow Prevention and Response* and *Infiltration/Inflow and Spill Prevention* requirements below.

Wastewater Collection System Overflow Prevention and Response

5. The Permittee shall coordinate with the appropriate local wastewater collection system entities on all relevant matters concerning the wastewater collection systems, pretreatment programs, and the wastewater treatment facility.
6. The Permittee shall minimize the discharge of chlorine, or any other toxic substance used for disinfection and cleanup of sewage overflows, to any surface water body. The Permittee shall take all reasonable steps to contain and prevent chlorine discharges to surface waters and minimize or correct any adverse impact on the environment resulting from the cleanup of overflows. The Permittee shall develop a monitoring program to evaluate the effectiveness of overflow cleanup protocols for protecting public health and the environment. Minimum protocols should include visual observation, sample collection, and sampling data analyses. The monitoring program shall be developed in coordination with the Regional Board and the Santa Barbara County Health Department. The Permittee shall submit a proposed monitoring program for Executive Officer review and approval **by March 1, 2005**.
7. The Permittee shall make every reasonable effort to prevent sewage overflows from its wastewater collection system and private systems from entering storm drains and/or surface water bodies. The Permittee shall also make every reasonable effort to prevent sewage and/or chlorine used for disinfection of overflows from discharging from storm drains into flood control channels and open ditches by blocking the storm drainage system and by removing the sewage and/or chlorine from the storm drains.

8. Upon reduction, loss, or failure of the wastewater collection system resulting in a sewage overflow, the Permittee shall, to the extent necessary to maintain compliance with this Order, take any necessary remedial action to:

- a. control or limit the volume of sewage discharged;
- b. terminate the sewage discharge as rapidly as possible, and;
- c. recover as much of the sewage discharged as possible for proper disposal, including any wash-down water.

The Permittee shall implement all remedial actions to the extent they may be applicable to the discharge, including the following:

- d. Interception and rerouting of sewage flows around the sewage line failure;
 - e. Vacuum truck recovery of wastewater collection system overflows and wash-down water;
 - f. Cleanup of debris of sewage origin at the overflow site;
 - g. Sample affected receiving water body to ensure adequate clean-up, and;
 - h. Submit monitoring data to the Executive Officer within 30 days of sampling.
9. The discharge of untreated or partially treated sewage is prohibited pursuant to Standard Provisions, Prohibition A.4, and shall constitute a violation of these discharge requirements unless the Permittee demonstrates through properly signed, contemporaneous operating logs, or other relevant evidence that the following criteria are met:
- a. The discharge was caused by one or more severe natural conditions, including hurricanes, tornadoes, widespread flooding, earthquakes, tsunamis, and other similar natural conditions, and;

- b. There were no feasible alternatives to the discharge, such as the use of auxiliary treatment facilities, retention of untreated wastewater, reduction of inflow and infiltration, use of adequate backup equipment, or an increase in the capacity of the system. This provision is not satisfied if, in the exercise of reasonable engineering judgment, the Permittee should have installed auxiliary or additional collection system components, wastewater retention or treatment facilities, or adequate back-up equipment, or should have reduced inflow and infiltration.

10. In any enforcement action, the Regional Board will consider the efforts of the Permittee to contain, control, and clean up sewage overflows from its collection system as part of the Board's consideration of the factors required by Section 13385 of the California Water Code.

Infiltration/Inflow and Spill Prevention Measures

11. The Permittee shall continue to develop and implement infiltration, inflow, and spill prevention efforts to address problems associated with infiltration (e.g., groundwater entering into the collection system through defective pipe joints or connections to manholes), inflow (e.g., storm water entering manhole covers) and sewage spills (often caused by grease or root blockages). These activities shall be reviewed and updated as necessary by **September 1st of every year**, and shall be incorporated into the Wastewater Collection System Management Plan as required by this Order, and as outlined in Attachment 1 to MRP No. R3-2004-0122. [See Sections IV.(E) and IX.(A) of MRP Attachment 1 for Infiltration/Inflow related requirements.]
12. Infiltration, inflow, and spill prevention measures shall be developed in accordance with good engineering practices and shall address the following objectives:
- a. Identify infiltration and inflow sources that may affect treatment facility operation or

possibly result in overflow or exceed pump station capacity; and,

- b. Identify, assign, and implement spill prevention measures and collection system management practices to ensure overflows and the contribution of pollutants (including illicit contributions) or "**incompatible wastes**" to the Discharger's treatment system are minimized.
13. Infiltration, inflow, and spill prevention measure documentation shall provide a description of the collection and transport system, measures used to ensure proper operation, and other information necessary to determine compliance with these requirements. The program shall include, at a minimum, the following items:
- a. A map showing collection system lines greater than 12 inches, pump stations, standby power facilities, surface water bodies (including discharge point(s) where pump station overflows may occur), storm drain inlets, and date of last revision.
 - b. A narrative description of the following:
 - 1) Line Flushing and Cleaning: Describe available equipment and projected schedule necessary to respond to the cleaning and flushing requirements for the collection system as necessary to prevent overflows, and assigned staff. Describe coordination with area plumbers to address introduction of "**incompatible wastes**" (e.g., root balls) during lateral cleaning, and efforts to abate introduction of materials (e.g., construction debris) into the system, which may cause system backup.
 - 2) Visual System Inspection: Describe visual inspection methods (e.g., televising lines), replacement schedules, inspection frequency, collection system length, and assigned staff. Describe results and provide details regarding problem areas
- detected. Inspection records shall be retained for five years.
- 3) Inflow & Infiltration: Describe current and five-year projected investigation methods (e.g., smoke testing), frequency, results, and efforts to reduce storm water inflows and sewer line exfiltration. Inspection records shall be retained for five years.
 - 4) Preventive Repair and Replacement: Describe a projected schedule to eliminate sewage conveyance systems determined or projected to be structurally compromised. Separately list each project or reach of conveyance to be replaced, along with proposed start and estimated completion dates.
 - 5) Pump Station Maintenance: Describe each pump station, location, flow monitoring (wet and dry weather), and the previous year's operational problems and overflows.
 - 6) Alternate Power Supply for Pump Station Operation: Describe alternate power supply for each pump station within the City's system.
14. Fiscal Resources: The City shall provide a description of fiscal resources necessary to ensure system operation. The description shall include, at a minimum, the following items:
- a. Fee Structure: Quantification of current and five-year projected sewer assessment fees necessary to implement the City's program, including a comparison of fees collected by the City with those collected by each local wastewater collection entity.
 - b. Available Fiscal Resources: Actual and five-year projected budget expenses for staffing, operation, and replacement of the collection system, including a description of a capital improvement or sinking fund to provide funding for item 16.e, below.

15. Personnel and Training: Infiltration, inflow, and spill prevention measures shall provide a description of staffing available to ensure system operation. The program shall include, at a minimum, the following items:
 - a. Personnel: Identify specific individuals (and job titles) who are responsible for developing, implementing, and revising the City's program. Provide an organizational chart of all staff, positions, duties, and training received during the past year. Identify managers and provide a list of contacts with associated telephone numbers.
 - b. Training: List the frequency of training, the qualifications of each employee, and coordination efforts between the City and local wastewater collection entities. Periodic dates for training shall be identified.
 16. Planning and Reporting: Infiltration, inflow, and spill prevention measures shall provide a description of planning efforts and reporting of system operation. The program shall include, at a minimum, the following items:
 - a. Spill Response: Describe a plan, and identify employees responsible and duties necessary to implement the Permittee's responses to spills. Identify posting, notification, and spill estimation practices used.
 - b. Annual Reporting (Due March 31st of each year; see Requirement D.17 below): List spills or system problems during the previous year, cleanups, amounts, locations, and corrective actions taken to ensure similar spills or problems do not recur. A tracking or follow-up procedure shall be used to ensure appropriate response has been taken. Inspections and maintenance activities shall be documented and recorded.
 - c. Offsite and Onsite Spill Alarms: Describe the current or proposed alarm system, central monitoring and information location, staffing, and response times for detecting spills from the system.
 - d. Wet Season Manhole Inspections: Describe or propose frequency to conduct inspections to detect line blockage during wet season flows and to avoid system overflows, staffing, and available and anticipated equipment to ensure safe and effective inspections.
 - e. Capital Improvement: Describe a current and projected work plan.
 - f. Five-Year Planning: Describe projected planning efforts.
 - g. 20-Year Planning: Describe long-term planning efforts.
 17. The City shall provide an annual report, due March 31st of each year, describing infiltration, inflow, and spill prevention measure development and permit compliance over the previous calendar year. The reports shall be of sufficient content as to enable the Regional Board to determine compliance with all requirements.
- ### E. PRETREATMENT SPECIFICATIONS
- A Pretreatment Program is a regulatory program administered by the Permittee that implements National Pretreatment Standards. These standards are promulgated by the USEPA in accordance with Section 307 (b) and (c) of the Federal Clean Water Act (CWA). This permit implements General Pretreatment Regulations of 40 CFR 403, latest revision.
1. The Permittee shall be responsible for the performance of all pretreatment requirements contained in 40 CFR 403 and shall be subject to enforcement actions, penalties, fines, and other remedies by the USEPA, or other appropriate parties, as provided in the CWA, as amended (33 USC 1251 et seq.). The Permittee shall implement and enforce its Approved Publicly Owned Treatment Works (POTW) Pretreatment Program. Implementation of the Permittee's Approved POTW Pretreatment Program is hereby made an enforceable condition of this Order and Permit. USEPA or the State may initiate enforcement action against the Permittee

or an industrial user for non-compliance with applicable standards and requirements as provided in the CWA.

2. The Permittee shall enforce the requirements promulgated under Sections 307 (b), (c), & (d), and 402 (b) of the CWA. The Permittee shall cause industrial users subject to Federal Categorical Standards to achieve compliance no later than the date specified in those requirements or, in the case of a new industrial user, upon commencement of the discharge.
3. The Permittee shall perform the pretreatment functions as required in 40 CFR Part 403, including, but not limited to:
 - a. Implement the necessary legal authorities as provided in 40 CFR 403.8 (f)(1);
 - b. Enforce the pretreatment requirements under 40 CFR 403.5 and 403.6;
 - c. Implement the programmatic functions as provided in 40 CFR 403.8 (f)(2); and,
 - d. Provide the requisite funding and personnel to implement the pretreatment program as provided in 40 CFR 403.8 (f)(3).

Monitoring shall be conducted in accordance with the Monitoring and Reporting Program (MRP) of this Order (see MRP Section X, *Pretreatment Program Reporting*). Also see Sections B and C of the Standard Provisions.

F. BIOSOLIDS REQUIREMENTS

(Note: Language in this section was provided by USEPA as standard language for use in NPDES permits. "Biosolids" refers to non-hazardous sewage sludge as defined in 40 CFR 503.9. Sewage sludge that is hazardous as defined in 40 CFR 261 must be disposed in accordance with the Resource Conservation and Recovery Act (RCRA). Sludge with PCB levels greater than 50 mg/kg must be disposed in accordance with 40 CFR 761.)

1. Management of all solids and sludge must comply with all requirements of CFR Parts 257, 258, 501, and 503, including all monitoring, record-keeping, and reporting requirements. Since the State of California, hence the Regional and State Boards, has not been delegated the authority by the USEPA to implement the biosolids program, enforcement of biosolids requirements of CFR Part 503 will occur under USEPA's jurisdiction at this time.
2. All biosolids generated by the permittee shall be used or disposed of in compliance with the applicable portions of:
 - a. 40 CFR 503: for biosolids which are land applied (placed on the land for the purpose of providing nutrients or conditioning the soil for crops or vegetation), placed in surface disposal sites (placed on the land at dedicated land disposal sites or monofills for the purpose of disposal), stored, or incinerated;
 - b. 40 CFR 258: for biosolids disposed in municipal solid waste landfills; and,
 - c. 40 CFR 257: for all biosolids use and disposal practices not covered under 40 CFR 258 or 503.

40 CFR 503 Subpart B (land application) applies to biosolids applied for the purpose of enhancing plant growth or for land reclamation. 40 CFR 503 Subpart C (surface disposal) applies to biosolids placed on the land for the purpose of disposal.

The Permittee is responsible for ensuring that all biosolids produced at its facility are used or disposed of in compliance with these regulations, whether the Permittee uses or disposes of the biosolids itself or transfers them to another party for further treatment, use, or disposal. The Permittee is responsible for informing subsequent preparers, applicators, and disposers of the requirements that they must meet under 40 CFR 257, 258, and 503.

3. Duty to mitigate: The Permittee shall take all reasonable steps to prevent or minimize any

- biosolids use or disposal in violation of applicable regulations and/or which has a likelihood of adversely affecting human health or the environment.
4. No biosolids shall be allowed to enter wetlands or other waters of the United States.
 5. Biosolids treatment, storage, use, or disposal shall not contaminate groundwater.
 6. Biosolids treatment, storage, use, or disposal shall not create a nuisance such as objectionable odors or flies.
 7. The Permittee shall assure that haulers transporting biosolids off site for treatment, storage, use, or disposal take all necessary measures to keep the biosolids contained.
 8. If biosolids are stored for over two years from the time they are generated, the Permittee must ensure compliance with all the requirements for surface disposal under 40 CFR 503 Subpart C, or must submit a written notification to USEPA with the information in Section 503.20(b), demonstrating the need for longer temporary storage.
 9. Any biosolids treatment, disposal, or storage site shall have facilities adequate to divert surface runoff from adjacent areas, to protect the site boundaries from erosion, and to prevent any conditions that would cause drainage from the materials at the site to escape from the site. Adequate protection is defined as protection from at least a 100-year storm and from the highest tidal stage that may occur.
 10. The discharge of biosolids shall not cause waste material to be in a position where it is, or can be, conveyed from the treatment and storage sites and deposited in the waters of the State.
 11. The Permittee shall design its pretreatment program local discharge limitations to achieve the metals concentration limits in 40 CFR 503.13 Table 3.
 12. Inspection and Entry: The USEPA, Regional Board, or an authorized representative thereof, upon the presentation of credentials, shall be allowed by the Permittee, directly or through contractual arrangements with their biosolids management contractors, to:
 - a. Enter upon all premises where biosolids produced by the Permittee are treated, stored, used, or disposed, either by the Permittee or by another party to whom the Permittee transfers the biosolids for treatment, storage, use, or disposal;
 - b. Have access to and copy any records that must be kept under the conditions of this permit or of 40 CFR 503, by the Permittee or by another party to whom the Permittee transfers the biosolids for further treatment, storage, use, or disposal, and;
 - c. Inspect any facilities, equipment (including monitoring and control equipment), practices, or operations used in the biosolids treatment, storage, use, or disposal by the Permittee or by another party to whom the Permittee transfers the biosolids for treatment, storage, use, or disposal.
 13. Monitoring shall be conducted in accordance with the Monitoring and Reporting Program (MRP) of this Order (see MRP Section VIII, *Biosolids Monitoring, Reporting, and Notification*);
 14. All the requirements of 40 CFR 503 and 23 CCR, Division 3, Chapter 15, and 27 CCR, Division 2 are enforceable by the USEPA and this Regional Board whether or not the requirements are stated in an NPDES permit or any other permit issued to the Permittee.
- ### G. PROVISIONS
1. Order No. 99-40, *Waste Discharge Requirements for the City of Santa Barbara El Estero Wastewater Treatment Plant and Local Sewering Entity, Santa Barbara County*, adopted by the Regional Board on September 8, 1999, is hereby rescinded. Order No. R3-2004-

0122 is effective as of the date its adoption by the Regional Board.

2. The Discharger shall comply with Monitoring and Reporting Program No. R3-2004-0122, or any amendments thereto, as ordered by the Executive Officer.
3. The Discharger shall comply with all of the attached *Standard Provisions and Reporting Requirements for National Pollutant Discharge Elimination System Permits*, dated January 1985 (Standard Provisions). Paragraph (a) of item E.1 of the Standard Provisions shall apply only if the bypass is for essential maintenance to assure efficient operation.
4. **This Order expires on October 22, 2009**, and the Permittee must file a complete Report of Waste Discharge in accordance with Title 23, Division 3, Chapter 9, of the California Code of Regulations, at least 180 days prior to that date (or **no later than April 22, 2009**), if the discharge will continue.
5. If toxicity monitoring shows a violation of toxicity limitations of this Order or a toxicity objective in Table B of the Ocean Plan, the Permittee shall increase the frequency of toxicity testing to once per week and submit the results within 15 days after each test to the Regional Board Executive Officer. If the discharge consistently exceeds toxicity effluent limitations as determined by the Executive Officer, the Permittee shall conduct a Toxicity Reduction Evaluation (TRE), which shall include all reasonable steps to identify the source(s) of toxicity. Once sources of toxicity are identified, the Permittee shall take all reasonable steps necessary to reduce toxicity to the required level.

The basis of the TRE shall be EPA's *Methods for Aquatic Toxicity Identification Evaluations: Phase I, Toxicity Characterization Procedures, 2nd Edition*, 1991b (EPA 600-6-91-003), *Methods for Aquatic Toxicity Identification Evaluations: Phase II, Toxicity Identification Procedures for Samples Exhibiting Acute and Chronic Toxicity*, 1993a (EPA 600-R-92-080), *Methods for Aquatic Toxicity Identification*

Evaluations: Phase III, Toxicity Confirmation Procedures for Samples Exhibiting Acute and Chronic Toxicity, 1993b (EPA 600-R-92-081), and *Toxicity Reduction Evaluation Guidance for Municipal Wastewater Treatment Plants* (EPA 833-B-99-002, August 1999, or revised editions.

The Permittee shall initiate a TRE according to the following schedule:

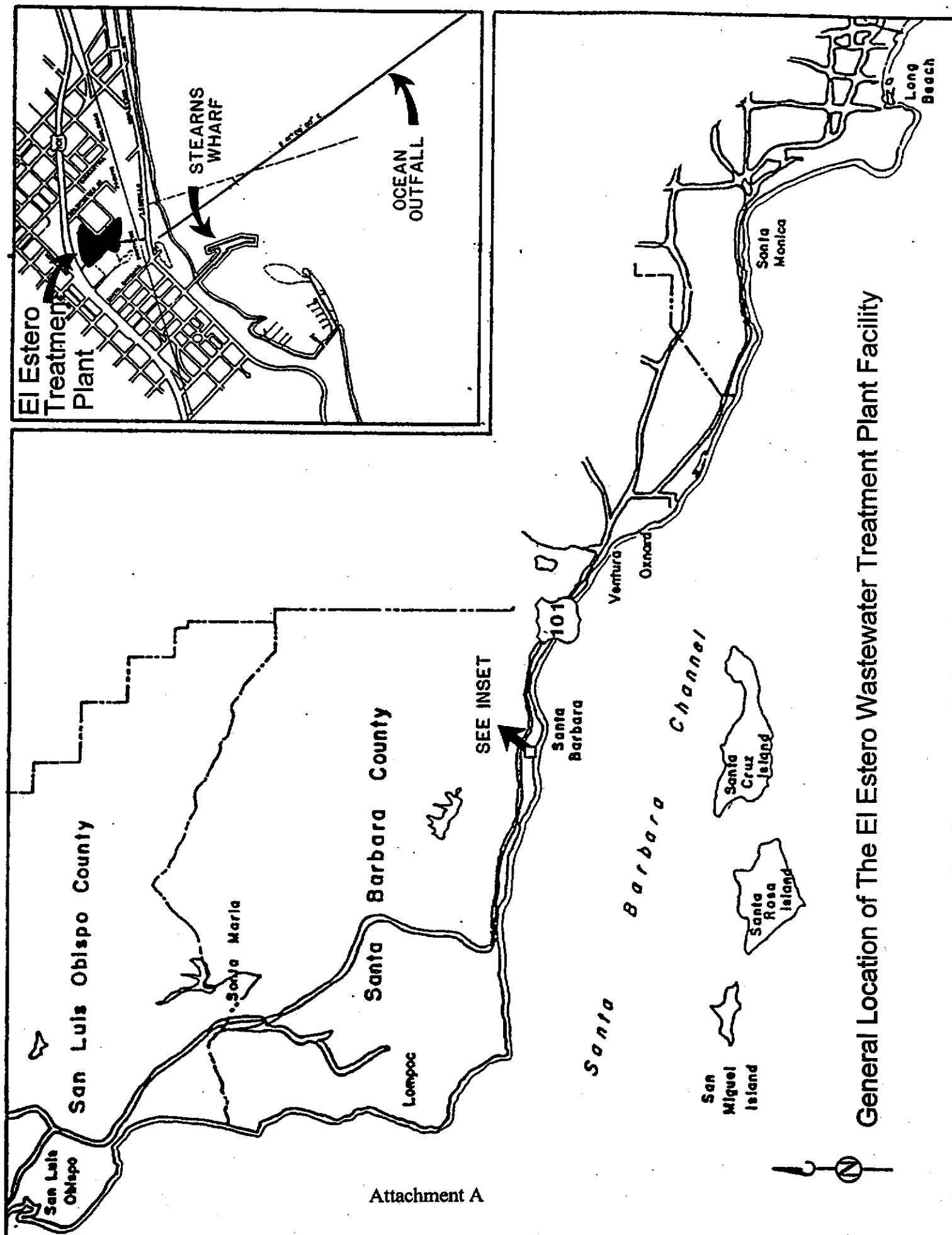
- a. Take all reasonable measures necessary to immediately reduce toxicity, where source is known [Within 24 hours of identification of noncompliance].
 - b. Submit to the Executive Officer a TRE study plan describing the toxicity reduction procedures to be employed [Within 60 days of identification of noncompliance].
 - c. Initiate the TRE [Time schedule to be determined by the Executive Officer].
 - d. Conduct the TRE following the procedures in the TRE study plan [Time schedule to be determined by the Executive Officer].
 - e. Submit the results of the TRE, including summary of findings, required corrective actions, and all results and data [Within 60 days of completing the TRE].
 - f. Implement corrective actions to meet permit limits and conditions [Within 7 days of notification by the Executive Officer].
 - g. Return to regular monitoring after implementing corrective measures and approval by the Executive Officer [One-year period or as specified in the TRE study plan].
6. If the projected waste brine and municipal discharge flowrates will vary from those specified in Finding No. 13, then the Permittee shall submit, for the approval of the Executive Officer, the results of computer modeling (also approved by the Executive Officer) to establish the required minimum initial dilution ratio for the combined discharge at the boundary of the zone of initial dilution. The Permittee shall

- submit the modeling results at least 60 days before proposing to begin the discharge of waste brine from the desalination plant per Provision G.7 (or, 240 days before the proposed date of discharge).
7. At least 180 days before the proposed date of discharge from the desalination plant, the Permittee shall inform the Executive Officer in writing. Regional Board staff may then draft revised waste discharge requirements establishing the revised minimum initial dilution ratio for the combined discharge and present them to the Board for their consideration at a regularly scheduled public meeting, or propose the appropriate revisions at the time of the next scheduled NPDES permit renewal.
 8. Waste management systems that discharge to the ocean must be designed and operated in a manner that will maintain the indigenous marine life and a healthy and diverse marine community^B.
 9. Waste effluents shall be discharged in a manner which provides sufficient initial dilution to minimize the concentrations of substances not removed in the treatment^B.
 10. Waste that contains pathogenic organisms or viruses should be discharged a sufficient distance from shellfishing and water-contact sports areas to maintain applicable bacterial standards without disinfection. Where conditions are such that an adequate distance cannot be attained, reliable disinfection in conjunction with a reasonable separation of the discharge point from the area of use must be provided. Disinfection procedures that do not increase effluent toxicity and that constitute the least environmental and human hazard should be used^B.
 11. The State Board is authorized to administer and enforce effluent limitations established pursuant to the Federal Clean Water Act. Effluent limitations established under Sections 301, 302, 306, 307, 316, 403, and 405 of the aforementioned Federal Act and administrative procedures pertaining thereto are included in the Ocean Plan by reference. Compliance with Ocean Plan Table A effluent limitations, or Environmental Protection Agency Effluent Limitations Guidelines for industrial discharges, based on Best Practicable Control Technology, shall be the minimum level of treatment acceptable under the Ocean Plan, and shall define reasonable treatment and waste control technology^B.

I, Roger W. Briggs, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Coast Region, on October 22, 2004.



Roger W. Briggs
Executive Officer



Attachment A

ATTACHMENT B

Effluent Limitations with Desalination Facility Brine Discharge

When the discharge consists of municipal wastewater with brine wastewater from the desalination facility, effluent shall not exceed the effluent limitations listed in Tables B-1b through B-3b (effluent limitations derived using a minimum initial seawater-to-effluent dilution ratio of 44 to 1).

TABLE B-1b – Effluent Limitations for the Protection of Marine Aquatic Life

	Units of Measurement	6-Month Median ⁴	Daily Maximum ⁵	Instantaneous Maximum ⁶
Arsenic	ug/L	230	1,300	3,500
Cadmium	ug/L	45	180	450
Chromium (Hexavalent) ¹	ug/L	90	360	900
Copper	ug/L	50	450	1,300
Lead	ug/L	90	360	900
Mercury	ug/L	1.8	7.2	18
Nickel	ug/L	220	900	2,200
Selenium	ug/L	680	2,700	6,800
Silver	ug/L	24	120	310
Zinc	ug/L	550	3,200	8,600
Cyanide ²	ug/L	45	180	450
Total Chlorine Residual ³	ug/L	90	360	2,700
Ammonia (expressed as N)	ug/L	27,000	110,000	270,000
Acute Toxicity	TUa	-----	1.6	-----
Chronic Toxicity	TUc	-----	45	-----
Phenolic Compounds (nonchlorinated)	ug/L	1,400	5,400	14,000
Chlorinated Phenolics	ug/L	45	180	450
Endosulfan	ug/L	0.40	0.81	1.2
Endrin	ug/L	0.090	0.18	0.27
HCH	ug/L	0.18	0.36	0.54
Radioactivity	Not to exceed limits specified in Title 17, Division 1, Chapter 5, Subchapter 4, Group 3, Article 3, Section 30253 of the California Code of Regulations. Reference to Section 30253 is prospective, including future changes to any incorporated provisions of federal law, as the changes take effect.			

¹ Dischargers may, at their option, meet this limitation as a total chromium limitation.^B

² If a Discharger can demonstrate to the satisfaction of the Regional Board (subject to USEPA approval) that an analytical method is available to reliably distinguish between strongly and weakly complexed cyanide, effluent limitations for cyanide may be met by the combined measurement of free cyanide, simple alkali metal cyanides, and weakly complexed organometallic cyanide complexes. In order for the analytical method to be acceptable, the recovery of free cyanide from metal complexes must be comparable to that achieved by the approved method in 40 CFR PART 136, as revised May 14, 1999.^B

³ Water quality objectives for total chlorine residual applying to intermittent discharges not exceeding two hours shall be determined using the following equation:

$$\log y = -0.43 (\log x) + 1.8$$

where: y = the water quality objective (in ug/L) to apply when chlorine is being discharged; and
x = the duration of uninterrupted chlorine discharge in minutes.

The applicable effluent limitation must then be determined using Equation No. 1 from the Ocean Plan.^B

- ⁴ The six-month median shall apply as a moving median of daily values for any 180-day period in which daily values represent flow weighted average concentrations within a 24-hour period. For intermittent discharges, the daily value shall be considered to equal zero for days on which no discharge occurred. The six-month median limit on daily mass emissions shall be determined using the six-month median effluent concentration as C_e and the observed flow rate Q in millions of gallons per day (each variable referring to Equation 3 of the Ocean Plan).^B
- ⁵ The daily maximum shall apply to flow weighted 24-hour composite samples. The daily maximum mass emission shall be determined using the daily maximum effluent concentration limit as C_e and the observed flow rate Q in millions of gallons per day (each variable referring to Equation 3 of the Ocean Plan).^B
- ⁶ The instantaneous maximum shall apply to grab sample determinations.^B

TABLE B-2b – Effluent Limitations for the Protection Of Human Health – Non-Carcinogens

Chemical	Units of Measurement	30-day average
Acrolein	ug/L	9.9×10^3
Antimony	ug/L	5.4×10^4
Bis(2-chloroethoxy) methane	ug/L	200
Bis(2-chloroisopropyl) ether	ug/L	5.4×10^4
Chlorobenzene	ug/L	2.6×10^4
Chromium (III)	ug/L	8.6×10^6
di-n-butyl phthalate	ug/L	1.6×10^5
Dichlorobenzenes	ug/L	2.3×10^5
Diethyl phthalate	ug/L	1.5×10^6
Dimethyl phthalate	ug/L	3.7×10^7
4,6-dinitro-2-methylphenol	ug/L	9.9×10^3
2,4-dinitrophenol	ug/L	180
Ethylbenzene	ug/L	1.9×10^5
Fluoranthene	ug/L	680
Hexachlorocyclopentadiene	ug/L	2.6×10^3
Nitrobenzene	ug/L	220
Thallium	ug/L	90
Toluene	ug/L	3.8×10^6
Tributyltin	ug/L	0.063
1,1,1-trichloroethane	ug/L	2.4×10^7

**TABLE B-3b – Effluent Limitations for the Protection Of Human Health –
Carcinogens**

Chemical	Units of Measurement	30-day average
Acrylonitrile	ug/L	4.5
Aldrin	ug/L	9.9×10^{-4}
Benzene	ug/L	270
Benzidine	ug/L	3.1×10^{-3}
Beryllium	ug/L	1.5
Bis(2-chloroethyl) ether	ug/L	2.0
Bis(2-ethylhexyl) phthalate	ug/L	160
Carbon tetrachloride	ug/L	40
Chlordane	ug/L	1.0×10^{-3}
Chlorodibromomethane	ug/L	390
Chloroform	ug/L	5.8×10^3
DDT	ug/L	7.7×10^{-3}
1,4-dichlorobenzene	ug/L	810
3,3'-dichlorobenzidine	ug/L	0.36
1,2-dichloroethane	ug/L	1.3×10^3
1,1-dichloroethylene	ug/L	40
Dichlorobromomethane	ug/L	280
Dichloromethane	ug/L	2.0×10^4
1,3-dichloropropene	ug/L	400
Dieldrin	ug/L	1.8×10^{-3}
2,4-dinitrotoluene	ug/L	120
1,2-diphenylhydrazine	ug/L	7.2
Halomethanes	ug/L	5.8×10^3
Heptachlor	ug/L	2.2×10^{-3}
Heptachlor epoxide	ug/L	9.0×10^{-4}
Hexachlorobenzene	ug/L	9.4×10^{-3}
Hexachlorobutadiene	ug/L	630
Hexachloroethane	ug/L	110
Isophorone	ug/L	3.3×10^4
N-nitrosodimethylamine	ug/L	330
N-nitrosodi-N-propylamine	ug/L	17
N-nitrosodiphenylamine	ug/L	110
PAHs	ug/L	0.40
PCBs	ug/L	8.6×10^{-4}
TCDD equivalents	ug/L	1.8×10^{-7}
1,1,2,2-tetrachloroethane	ug/L	100
Tetrachloroethylene	ug/L	90
Toxaphene	ug/L	9.4×10^{-3}
Trichloroethylene	ug/L	1.2×10^3
1,1,2-trichloroethane	ug/L	420
2,4,6-trichlorophenol	ug/L	13
Vinyl chloride	ug/L	1.6×10^3

ATTACHMENT C

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CONTACT INFORMATION

Executive Officer North Coast Regional Water Quality Control Board (1) 5550 Skylane Boulevard, Suite A Santa Rosa, CA 95403 Ph (707) 576-2220, FAX (707) 523-0135	Executive Officer San Francisco Bay Regional Water Quality Control Board (2) 1515 Clay Street, Suite 1400 Oakland, CA 94612 Ph (510) 622-2300, FAX (510) 622-2460
Executive Officer Central Coast Regional Water Quality Control Board (3) 895 Aerovista Place, Suite 101 San Luis Obispo, CA 93401 Ph (805) 549-3147, FAX (805) 543-0397	Executive Officer Los Angeles Regional Water Quality Control Board (4) 320 West 4 th Street, Suite 200 Los Angeles, CA 90013 Ph (213) 576-6600, FAX (213) 576-6640
Executive Officer Central Valley Regional Water Quality Control Board, Sacramento Branch Office (5S) 11020 Sun Center Drive, Suite 200 Rancho Cordova, CA 95670-6114 Ph (916) 464-3291, FAX (916) 464-4645	Assistant Executive Officer Central Valley Regional Water Quality Control Board, Fresno Branch Office (5F) 1685 E Street Fresno, CA 93706 Ph (559) 445-5116, FAX (559) 445-5910
Assistant Executive Officer Central Valley Regional Water Quality Control Board, Redding Branch Office (5R) 415 Knollcrest Drive Redding, CA 96002 Ph (530) 224-4845, FAX (530) 224-4857	Executive Officer Lahontan Regional Water Quality Control Board, South Lake Tahoe Office (6SLT) 2501 Lake Tahoe Boulevard South Lake Tahoe, CA 96150 Ph (530) 542-5400, FAX (530) 544-2271
Executive Officer Lahontan Regional Water Quality Control Board, Victorville Office (6V) 15428 Civic Drive, Suite 100 Victorville, CA 92392 Ph (760) 241-6583, FAX (760) 241-7308	Executive Officer Colorado River Basin Regional Water Quality Control Board (7) 73-720 Fred Waring Drive, Suite 100 Palm Desert, CA 92260 Ph (760) 346-7491, FAX (760) 341-6820
Executive Officer Santa Ana Regional Water Quality Control Board (8) 3737 Main Street, Suite 500 Riverside, CA 92501-3339 Ph (909) 782-4130, FAX (909) 781-6288	Executive Officer San Diego Regional Water Quality Control Board (9) 9174 Sky Park Court, Suite 100 San Diego, CA 92124-1324 Ph (858) 467-2952, FAX (858) 571-6972